

UCE825: LIFE CYCLE ASSESSMENT				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To introduce the concepts of circular economy and Life Cycle decision making for minimal environmental impact in manufacturing, commerce and service industries. Life Cycle analysis and Life Cycle Assessment will be applied to resource selection, process choices and waste management.				
Introduction to life cycle analysis (LCA): Boundaries, functional unit, life cycle assessment (LCA assessment) Concepts of LCA: Policy and organisational drivers, and application in practice life cycle assessment methodology including (e.g. stages of LCA, Setting the goal, scope, functional unit and boundaries of LCA, comparing products, identifying environmental indicators, interpretation of the LCA assessment (characterised vs normalised) Assessment Methods: International standards on LCA (e.g. ISO 14044, PAS 2050), Tools used for LCA (e.g. Open LCA), Principles and practice of design for sustainability. LCA in Construction: Environmental purchasing based on LCA, Reuse, recycle, recover, waste management, Symbiosis – one company’s waste becoming another company’s feedstock. Case Study: Undertake an LCA analysis for an activity/product/service, identify information inputs, environmental impacts and LCA analysis outputs, develop an assessment methodology to enable you to evaluate the relative impacts of the most significant aspects of all the potential solutions, and hence reach a conclusion of the preferred solution (the Life Cycle Assessment)				
Course Learning Objectives (CLO) The students will be able to: 1. Evaluate the economic, environmental, social and technical aspects that influence a manufacturing organisation 2. Appraise the product lifecycle and the related impact on a business 3. Analyse the information required to undertake an LCA 4. Undertake an LCA and assess its limitations.				
Text Books 1. Arvanitoyannis, I.S., 2008. ISO 14040: life cycle assessment (LCA)–principles and guidelines. <i>Waste management for the food industries</i>, pp.97-132. 2. Life Cycle Analysis in Construction industry, George Xanthakis, 2014 3. Life Cycle Assessment in the Built Environment, Robert Crawford, 2011				

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25